CLASSIFICATION

COMPARING CLASSIFIERS

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COMPARING CLASSIFIERS

The following concepts will be introduced:

- ✓ **CONFIDENCE INTERVAL**
- ✓ **EMPIRICAL ACCURACY**
- ✓ **HYPOTHESIS TESTING**
  - DIFFERENT TEST SETS
  - SAME TEST SET

Comparing Classifiers

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	n	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	y	7	1.78	OH	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	n	6	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	510	355-9993
147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173

You want to use the **Customers Calls Database**, provided by the **CEO of the Telecom Company**, to develop different **Classification Techniques** to **find** which is **the best classification model for solving the Churn Problem**.

However, the **concept of best classification model is difficult to define** and you know that what is needed to start is to **compare the Accuracy achieved by different Classification Models from different Classification Techniques**.

Comparing Classifiers

Indeed, you know that it is often useful to **COMPARE THE PERFORMANCE OF** different **CLASSIFICATION MODELS** to determine **WHICH ONE WORKS BETTER ON A GIVEN DATA SET**.

Depending on the size of the data, the **OBSERVED DIFFERENCE IN ACCURACY BETWEEN TWO CLASSIFICATION MODELS MAY NOT BE STATISTICALLY SIGNIFICANT**.

Consider the following situation

INDUCER A **Accuracy = 0.85** when tested on a test data set containing **30 records**

INDUCER B **Accuracy = 0.75** when tested on a test data set containing **5,000 records**

IS INDUCER A BETTER THAN INDUCER B?

Comparing Classifiers

Two **key questions regarding** the **Statistical Significance** of the performance metrics are raised:

Although **Inducer A** has a higher accuracy than **Inducer B**, it was tested on a smaller test set.

How much confidence can we place on the Accuracy for Inducer A?

Issue of estimating the **Confidence Interval of the Accuracy for Inducer A or B**.

Is it possible to explain the difference in Accuracy as a result of variations in the composition of the test sets?

Issue of **Testing the Statistical Significance of the observed deviation**.

We consider the task of **PREDICTING THE VALUE OF THE CLASS ATTRIBUTE FOR A TEST RECORD** as a **BINOMIAL EXPERIMENT**.

Given a **TEST SET D_N** containing **N RECORDS**, we let

X; number of records correctly predicted by a given Inducer

p; true, but unknown, Accuracy of the Inducer

By modelling the **PREDICTION TASK** as a **BINOMIAL EXPERIMENT**, X has a **BINOMIAL DISTRIBUTION** with

$$\text{mean} = N \cdot p \qquad \text{variance} = N \cdot p \cdot (1 - p)$$

The empirical Accuracy

$$acc = \frac{X}{N}$$

has a Binomial distribution with

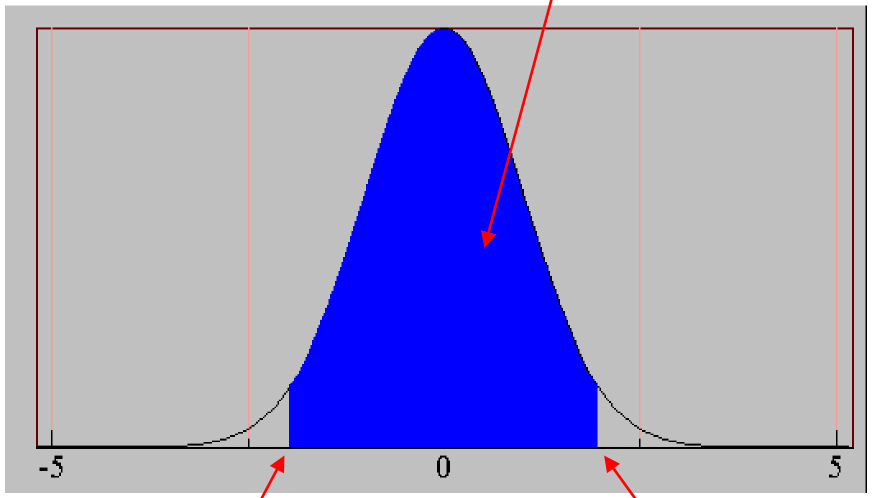
$$\text{mean} = p \qquad \text{variance} = p \cdot (1 - p)/N$$

Although the **BINOMIAL DISTRIBUTION** can be used to estimate the **CONFIDENCE INTERVAL FOR *acc***, it is often **APPROXIMATED BY A NORMAL DISTRIBUTION** when the number of records of the Test Set is sufficiently large.

Based on the **NORMAL DISTRIBUTION**, the following **CONFIDENCE INTERVAL FOR THE EMPIRICAL ACCURACY *acc*** is obtained:

$$P\left(-Z_{1-\frac{\alpha}{2}} < \frac{acc - p}{\sqrt{p \cdot (1-p)/N}} < Z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

Area = 1 - α



$-Z_{1-\frac{\alpha}{2}}$ $\qquad$ $Z_{1-\frac{\alpha}{2}}$

Although the **Binomial Distribution** can be used to estimate the **Confidence Interval for *acc***, it is often **approximated by a Normal Distribution** when the number of records of the Test Set is sufficiently large.

Based on the **Normal Distribution**, the following **Confidence Interval for the empirical Accuracy *acc*** is obtained:

$$P\left(-Z_{1-\frac{\alpha}{2}} < \frac{acc - p}{\sqrt{p\cdot(1-p)/N}} < Z_{1-\frac{\alpha}{2}}\right) = 1-\alpha$$

Rearranging the above inequality leads to the following **Confidence Interval with significance α (confidence $1-\alpha$) for p,** the unknown **Classification Model's Accuracy**

$$\left(\frac{acc + \frac{Z^2_{1-\frac{\alpha}{2}}}{2\cdot N} - Z_{1-\frac{\alpha}{2}}\cdot\sqrt{\frac{acc}{N} - \frac{acc^2}{N} + \frac{Z^2_{1-\frac{\alpha}{2}}}{4\cdot N^2}}}{\left(1+\frac{Z^2_{1-\frac{\alpha}{2}}}{N}\right)}, \frac{acc + \frac{Z^2_{1-\frac{\alpha}{2}}}{2\cdot N} + Z_{1-\frac{\alpha}{2}}\cdot\sqrt{\frac{acc}{N} - \frac{acc^2}{N} + \frac{Z^2_{1-\frac{\alpha}{2}}}{4\cdot N^2}}}{\left(1+\frac{Z^2_{1-\frac{\alpha}{2}}}{N}\right)}\right)$$

Given a **CLASSIFICATION MODEL** achieving an **EMPIRICAL ACCURACY *acc*=0.8** on a **TEST SET** consisting of **100 records (N=100)** we want to know the

CONFIDENCE INTERVAL for its true ACCURACY at a 95% confidence level

The 95% confidence level corresponds to

$$Z_{0.975} = 1.96$$

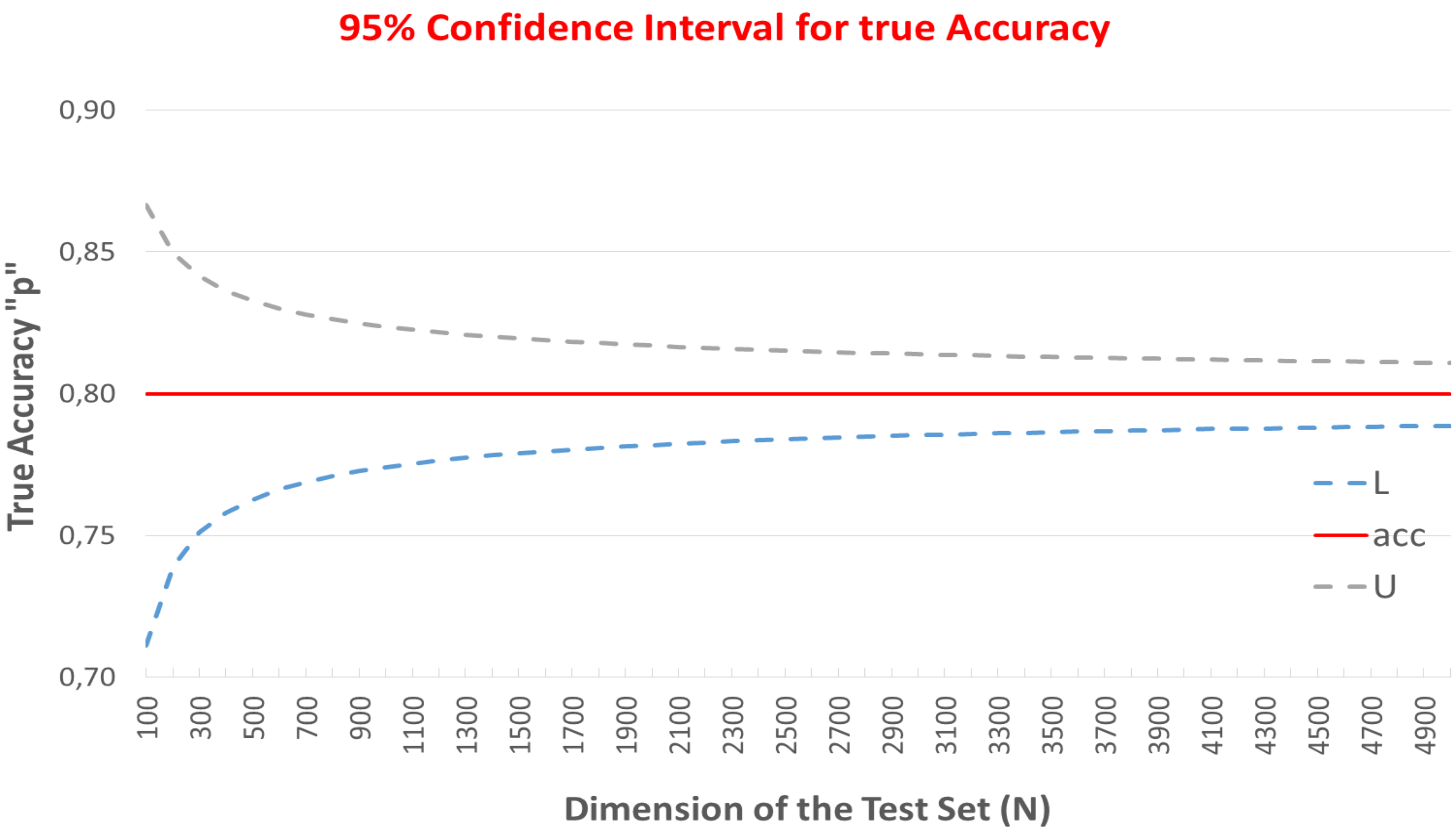
which brings to the following **CONFIDENCE INTERVAL FOR THE TRUE UNKNOWN ACCURACY OF THE CONSIDERED CLASSIFICATION MODEL**

$$(0.711, 0.867)$$

Wilson score interval (Wilson, 1927)

$$\left(\frac{acc + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{2 \cdot N} - Z_{1-\frac{\alpha}{2}} \cdot \sqrt{\dfrac{acc}{N} - \dfrac{acc^2}{N} + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{4 \cdot N^2}}}{\left(1 + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{N}\right)}, \frac{acc + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{2 \cdot N} + Z_{1-\frac{\alpha}{2}} \cdot \sqrt{\dfrac{acc}{N} - \dfrac{acc^2}{N} + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{4 \cdot N^2}}}{\left(1 + \dfrac{Z^2_{1-\frac{\alpha}{2}}}{N}\right)} \right)$$

What happens when the number of records N increases?



COMPARING CLASSIFIERS: DIFFERENT TEST SETS

Consider two Classification Models

- **M_1** evaluated on the Test Set **D_1** containing **n_1** records achieves Error rate **e_1**
- **M_2** evaluated on the Test Set **D_2** containing **n_2** records achieves Error rate **e_2**

The two Test Sets D_1 and D_2 are assumed to be independent.

Our goal is to test whether the observed difference between **e_1** and **e_2** is **STATISTICALLY SIGNIFICANT**.

Assume that **n_1 and n_2 are sufficiently large**, the **Error Rates e_1 and e_2 can be approximated using Normal Distributions**.

If the **Observed Difference in the Error Rate** is denoted as

$$d = e_1 - e_2$$

it is also **distributed according to a Normal Distribution** with

$$\text{mean} = d_t \qquad \text{variance} = \sigma_d^2$$

The variance of **d** can be estimated as follows:

$$\sigma_d^2 \cong \hat{\sigma}_d^2 = \frac{e_1 \cdot (1 - e_1)}{n_1} + \frac{e_2 \cdot (1 - e_2)}{n_2}$$

The **Confidence Interval for the true difference d_t** is

$$\boxed{\left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d , d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

$$\left(d - z_{1-\alpha/2} \cdot \widehat{\sigma}_d, d + z_{1-\alpha/2} \cdot \widehat{\sigma}_d\right)$$

IF THE CONFIDENCE INTERVAL SPANS THE VALUE 0, WE CONCLUDE THAT THE OBSERVED DIFFERENCE d IS NOT STATISTICALLY SIGNIFICANT AT THE α LEVEL (CONFIDENCE 1-α).

$$\boxed{0 \in \left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d, d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

$$\boxed{\left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d, d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

Classification models M_1 and M_2 are not significantly different in terms of achieved error at the α level (confidence $1-\alpha$).

If the confidence interval spans the value 0, we conclude that the observed difference d is not statistically significant at the α level (confidence $1-\alpha$).

$$\boxed{0 \in \left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d, d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

$$\boxed{\left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d, d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

IF THE UPPER LIMIT OF THE CONFIDENCE INTERVAL IS NEGATIVE THEN THE CLASSIFICATION MODEL M_1 IS BETTER THAN THE CLASSIFICATION MODEL M_2 AT THE $\alpha/2$ LEVEL (CONFIDENCE $1-\alpha/2$).

$$\boxed{d + z_{1-\alpha/2} \cdot \hat{\sigma}_d < 0}$$

$$\uparrow$$

$$\boxed{\left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d , d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)}$$

IF THE LOWER LIMIT OF THE CONFIDENCE INTERVAL IS POSITIVE THEN THE CLASSIFICATION MODEL M_2 IS BETTER THAN THE CLASSIFICATION MODEL M_1 AT THE $\alpha/2$ LEVEL (CONFIDENCE $1-\alpha/2$).

$$d - z_{1-\alpha/2} \cdot \hat{\sigma}_d > 0$$

$$\left(d - z_{1-\alpha/2} \cdot \hat{\sigma}_d , d + z_{1-\alpha/2} \cdot \hat{\sigma}_d\right)$$

COMPARING CLASSIFIERS: SAME TEST SET

Consider two Classification Models **M_1 AND M_2 TO BE COMPARED USING THE K-FOLDS CROSS VALIDATION PROCEDURE (KF-CV)**.

The dataset **D** is partitioned into **K** disjoint subsets, exhaustive (a partition of **D**) and with almost a constant number of records

$$D_1, D_2, \ldots, D_K$$

We then **APPLY CLASSIFICATION TECHNIQUES TO CONSTRUCT CLASSIFICATION MODELS M_1** and **M_2** from **K-1** of the partitions and test them on the remaining partition. This step is **REPEATED K TIMES**, each time using a different partition as the test set.

Let

- **M_{1k}** inducer for model **M_1** obtained at the **K-TH ITERATION** (k=1, …, K)
- **M_{2k}** inducer for model **M_2** obtained at the **K-TH ITERATION** (k=1, …, K)

Each pair of models **M_{1k}** and **M_{2k}** are **TESTED ON THE SAME PARTITION K**.

Comparing Classifiers: Same Test Set

Let

- **e_{1k}** error rate of **M_{1k}**
- **e_{2k}** error rate of **M_{2k}**

The **difference between their error rates during the k-th iteration** (fold) can be written as:

$$d_k = e_{1k} - e_{2k}$$

If **K** is **sufficiently large**, then **d_k** is **Normally Distributed** with:

$$\text{mean} = d_t^{cv} \qquad \text{standard deviation} = \sigma^{cv}$$

The **Overall Variance in the Observed Differences** is estimated using the following formula:

$$\hat{\sigma}^2_{d^{cv}} = \frac{\sum_{k=1}^{K}\left(d_k - \bar{d}\right)^2}{K \cdot \left(K - 1\right)} \qquad \bar{d} = \frac{1}{K}\sum_{k=1}^{K} d_k$$

We use the **STUDENT T DISTRIBUTION** to compute the **CONFIDENCE INTERVAL FOR THE VALUE OF THE TRUE MEAN d_t^{cv}**

$$\boxed{\left(\bar{d} - t_{1-\alpha/2}^{K-1} \cdot \hat{\sigma}_{d^{cv}}, \bar{d} + t_{1-\alpha/2}^{K-1} \cdot \hat{\sigma}_{d^{cv}}\right)}$$

where

$$t_{1-\alpha/2}^{K-1}$$

is obtained from a probability table, we get the quantile associated with **CONFIDENCE $1-\alpha$ AND K-1 DEGREES OF FREEDOM**.

Concerning the use of the **CONFIDENCE INTERVAL TO ESTABLISH STATISTICAL SIGNIFICANCE**, the same considerations made for the case of **2 CLASSIFIERS** apply here.

IF THE CONFIDENCE INTERVAL SPANS THE VALUE 0, WE CONCLUDE THAT THE OBSERVED DIFFERENCE IS NOT STATISTICALLY SIGNIFICANT AT THE α LEVEL (CONFIDENCE $1-\alpha$).